

Erdős Problem #575

EP-575

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

If $\mathcal{F}$ is a finite set of finite graphs then $\text{ex}(n; \mathcal{F})$ is the maximum number of edges a graph on n vertices can have without containing any subgraphs from $\mathcal{F}$. Note that it is trivial that $\text{ex}(n; \mathcal{F}) \leq \text{ex}(n; G)$ for every $G \in \mathcal{F}$.

Is it true that, for every $\mathcal{F}$, if there is a bipartite graph in $\mathcal{F}$ then there exists some bipartite $G \in \mathcal{F}$ such that $\text{ex}(n; G) \ll_{\mathcal{F}} \text{ex}(n; \mathcal{F})$?